MBB: ORIGINAL SOLID KEPT

```
vshell run summary: MBB
================================================================================

decision
--------
  outcome                         original SOLID part kept
  reason                          shelling does not improve the part: combined
                                  gain J = +0.0003 is not above ln(1 + 0.02)
  best design (k solid regions)   27
  final mesh                      final_design.stl

part
----
  STEP file                       /home/claude/vshell_repo/examples/step/Solid
                                  MBB.step
  volume (B-rep) mm3              8.538e+04
  volume (field) mm3              8.528e+04
  mass g                          88.79
  faces                           192
  analytic faces                  76
  support faces                   [10, 20, 106, 116]
  load faces                      [[34, 47, 130, 143]]
  force N                         [0, 0, -200]

segmentation
------------
  joints                          12
  struts                          16
  mounts                          8
  rank order (most shear-critical first) [22, 11, 1, 21, 23, 2, 18, 12, 10, 9, 7, 6, 4,
                                  3, ...]

baseline (solid)
----------------
  stiffness K0 N/mm               472.9
  load capacity P0 N              2742
  governing mode                  strength
  lambda strength                 13.71
  lambda Euler                    39.73
  von Mises q MPa                 2.67
  mount von Mises q MPa           5.316
  FE DOF                          701595
  FE cells                        192134
  CG iterations                   35

best hollow candidate
---------------------
  solid regions                   [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
                                  14, ...]
  hollow regions                  [26]
  outer offset / strut wall s mm 1.765
  as-built wall mm                1.565
  stiffness K N/mm                473.1
  K/K0                            1
  load capacity P N               2743
  P/P0                            1
  J                               3.4998e-04
  governing mode                  strength
  governing region                -
  lambda strength / Euler / local [13.72, 39.74, 1541]
  mass (field) g                  88.69
  enclosed voids                  1
  filled small voids mm3          0
  mesh volume mm3                 8.512e+04
  mesh watertight                 no

search
------
  hollow designs analysed         10
  feasible in mass sweep          28
  objective                       {'w_stiffness': 0.5, 'w_capacity': 0.5,
                                  'min_gain': 0.02, 'floor_stiffness': 0.85,
                                  'floor_capacity': 0.85}

grid
----
  sdf_mm                          0.4
  shape                           [776, 300, 152]
  fe_mm                           0.8
  seg_mm                          1
```

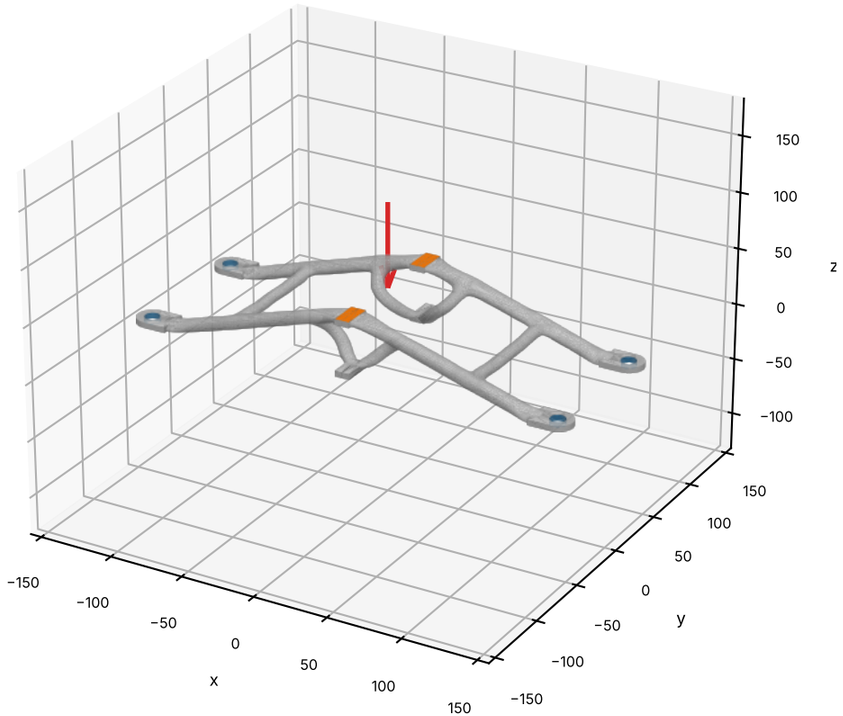
```
  n_exact                          5810241
  n_active                         4680370

timings s
---------
  geometry                         4.192
  sdf                              211.6
  segmentation                     13.02
  fe                               884.2
  mass_sweep                       160.5
  outputs                          7.308
  total                            1362

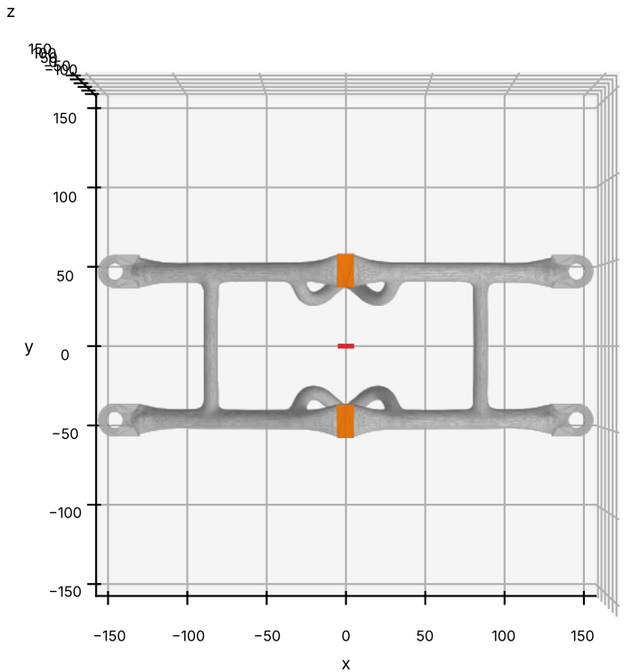
provenance
----------
  time (UTC)                       2026-10-28T08:40:04+00:00
  host                             vm
  platform                         Linux-6.18.44-fc-v180-x86_64-with-glibc2.39
  python                           3.13.18
  git                              -
  STEP sha256                      04fc4ba2900a0fcd
  packages                         numpy 2.5.3, scipy 1.18.1, scikit-image
                                   0.26.0, skan 0.13.1, libigl 2.6.3, pyamg
                                   5.3.0, trimesh 5.2.0, matplotlib 3.11.2,
                                   build123d 0.13.0
```

MBB: supports (blue, all DOF fixed) and load (orange faces, red arrow, |F| = 200 N)

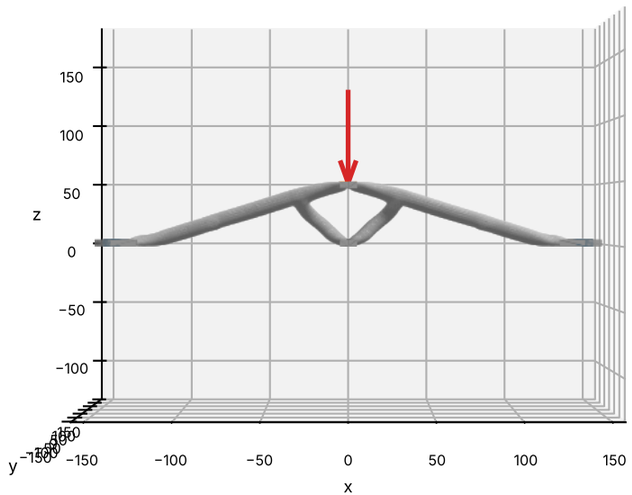
isometric



top (+z)

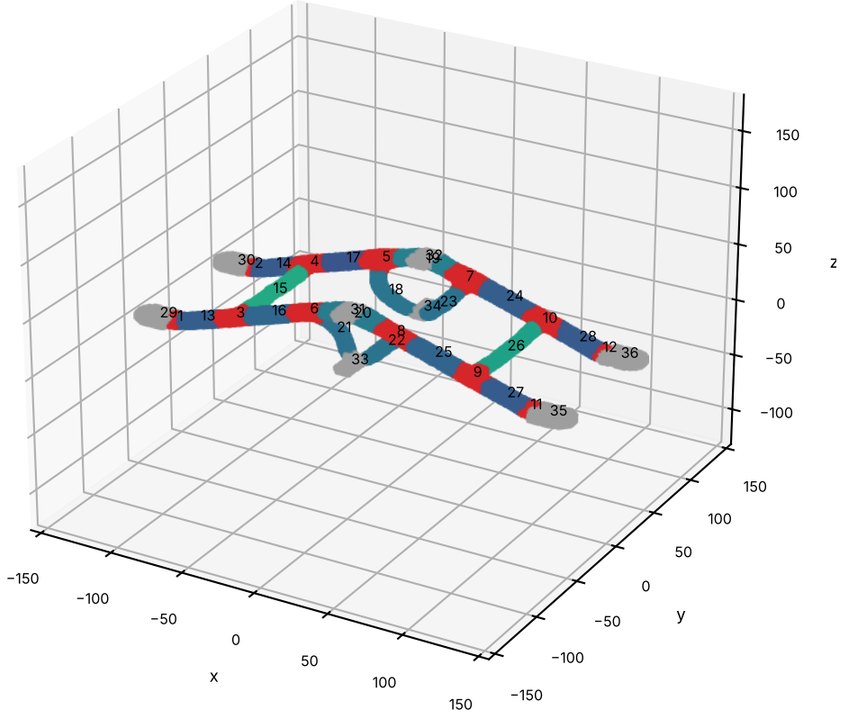


front (-y)

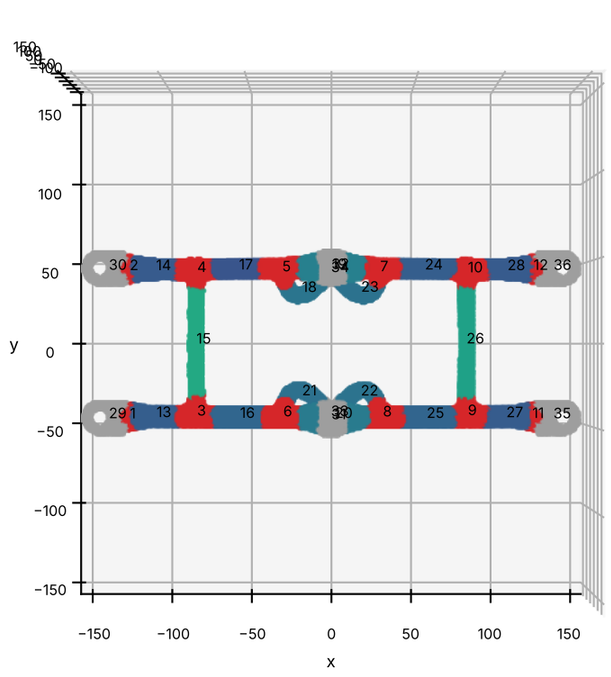


Segmentation: 12 joints (red), 16 struts (colour = slenderness lambda, 0 to 60+), 8 mounts (grey, never hollowed); numbers = region ids

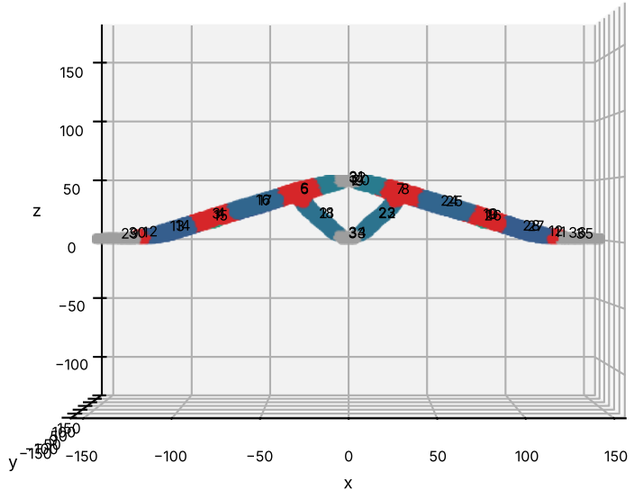
isometric



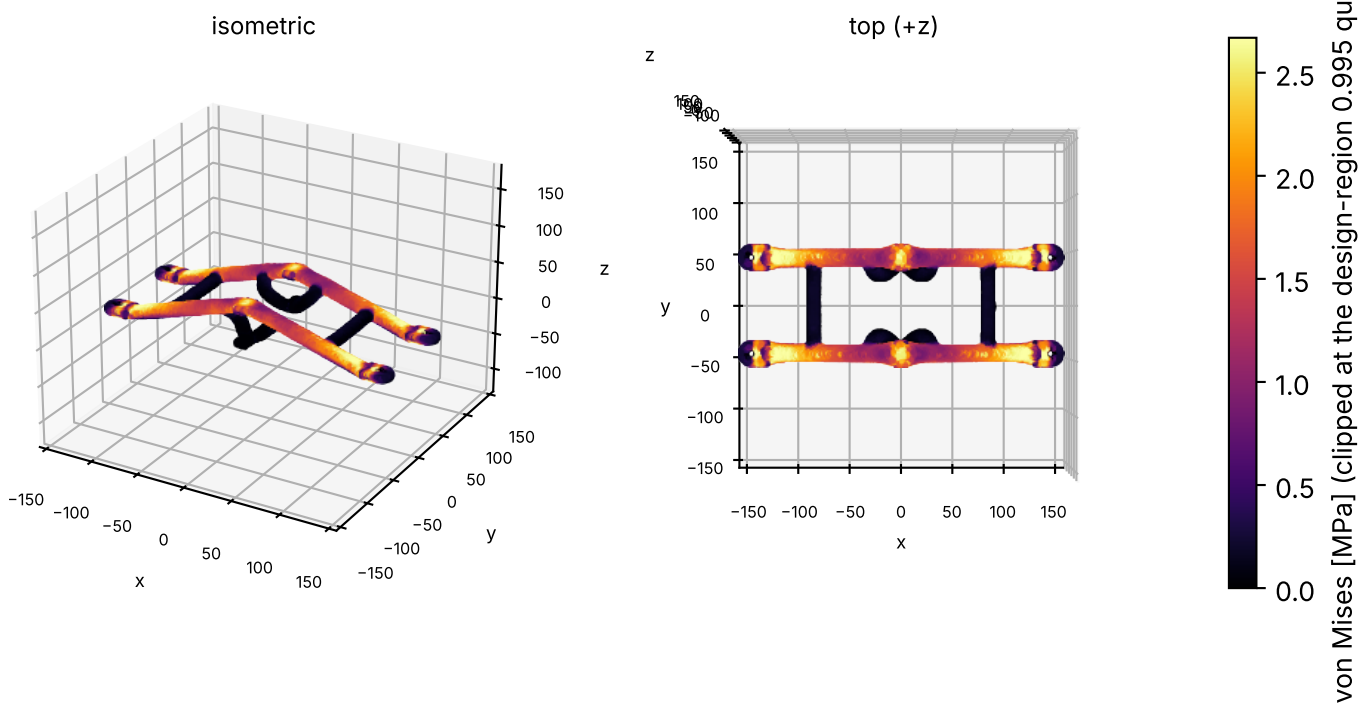
top (+z)

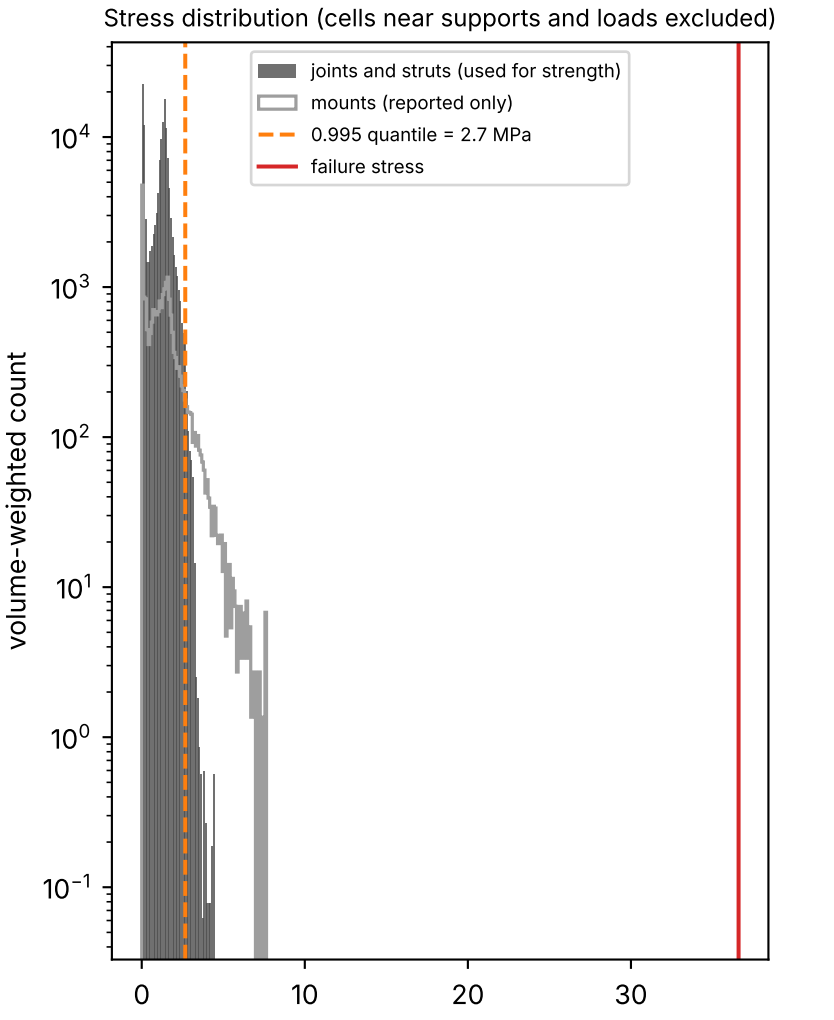


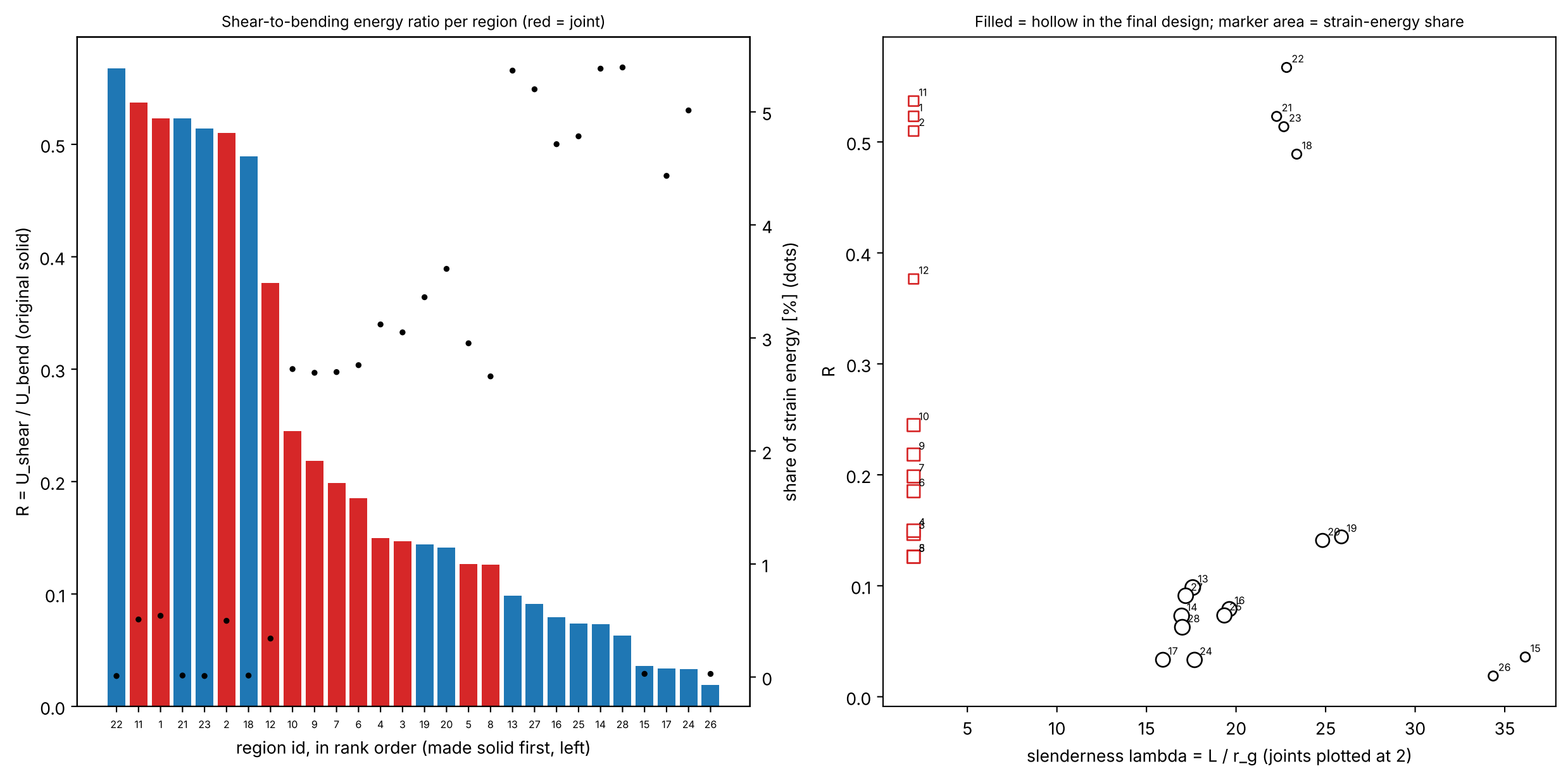
front (-y)

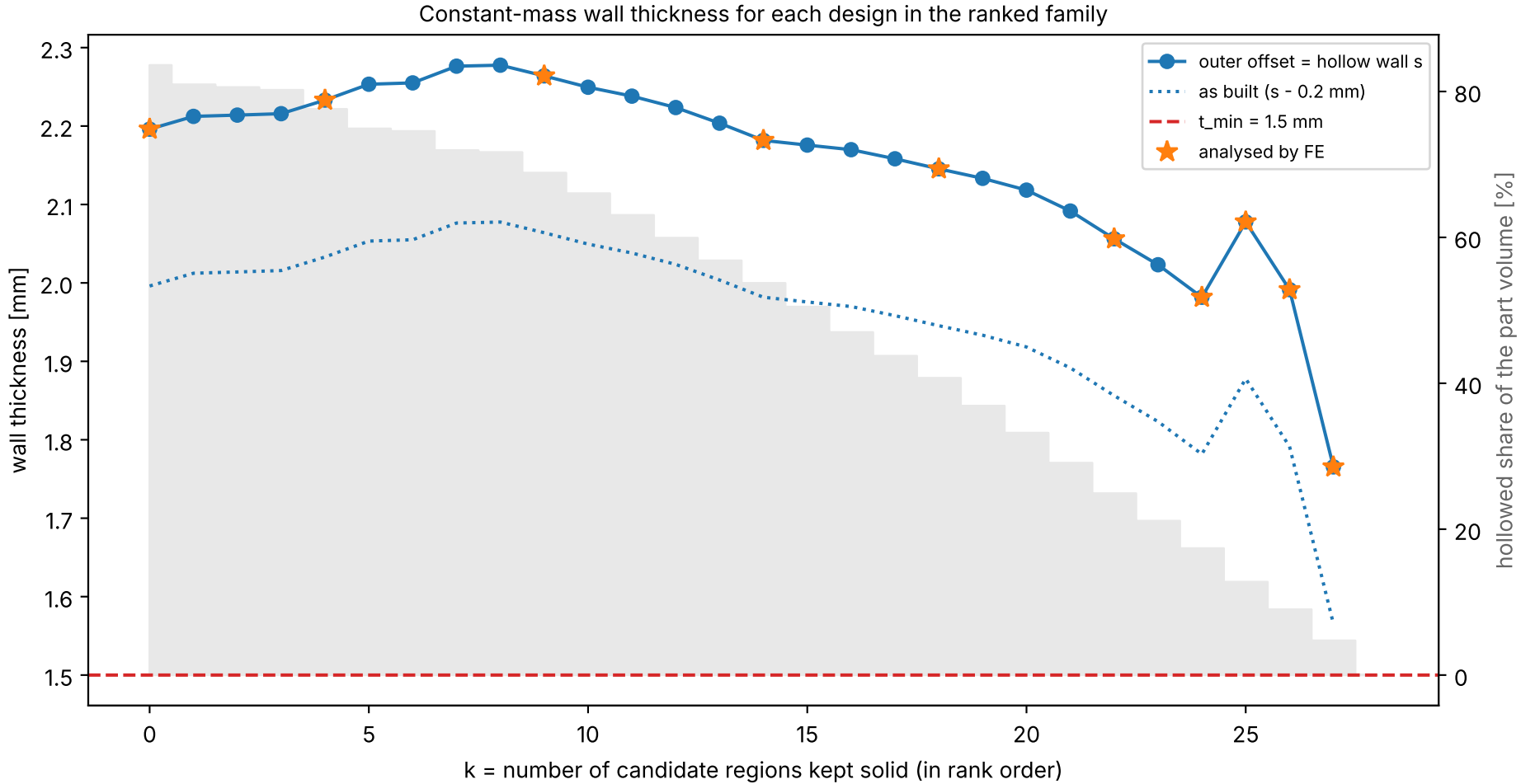


Original solid: von Mises stress at |F| = 200 N



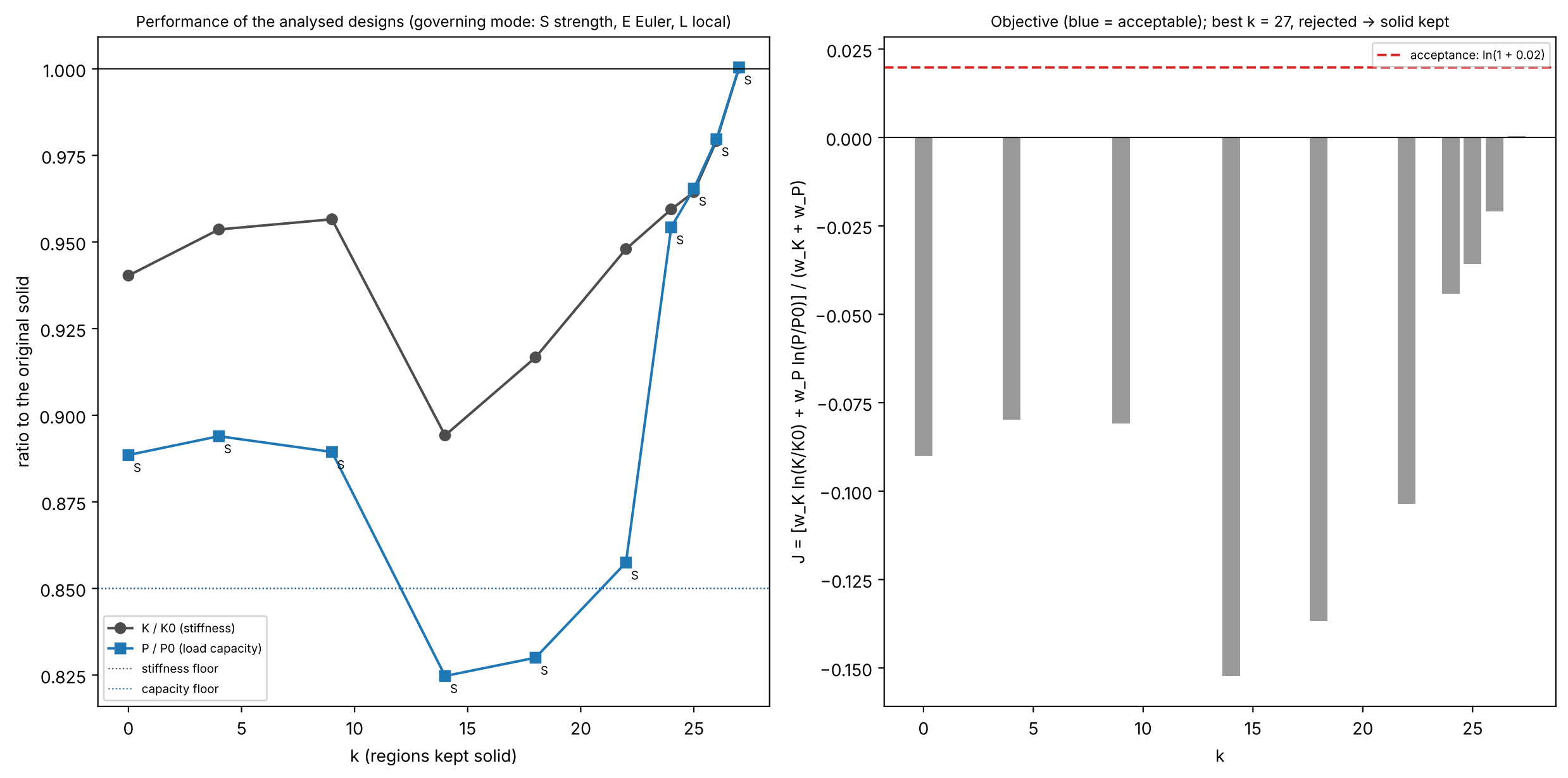


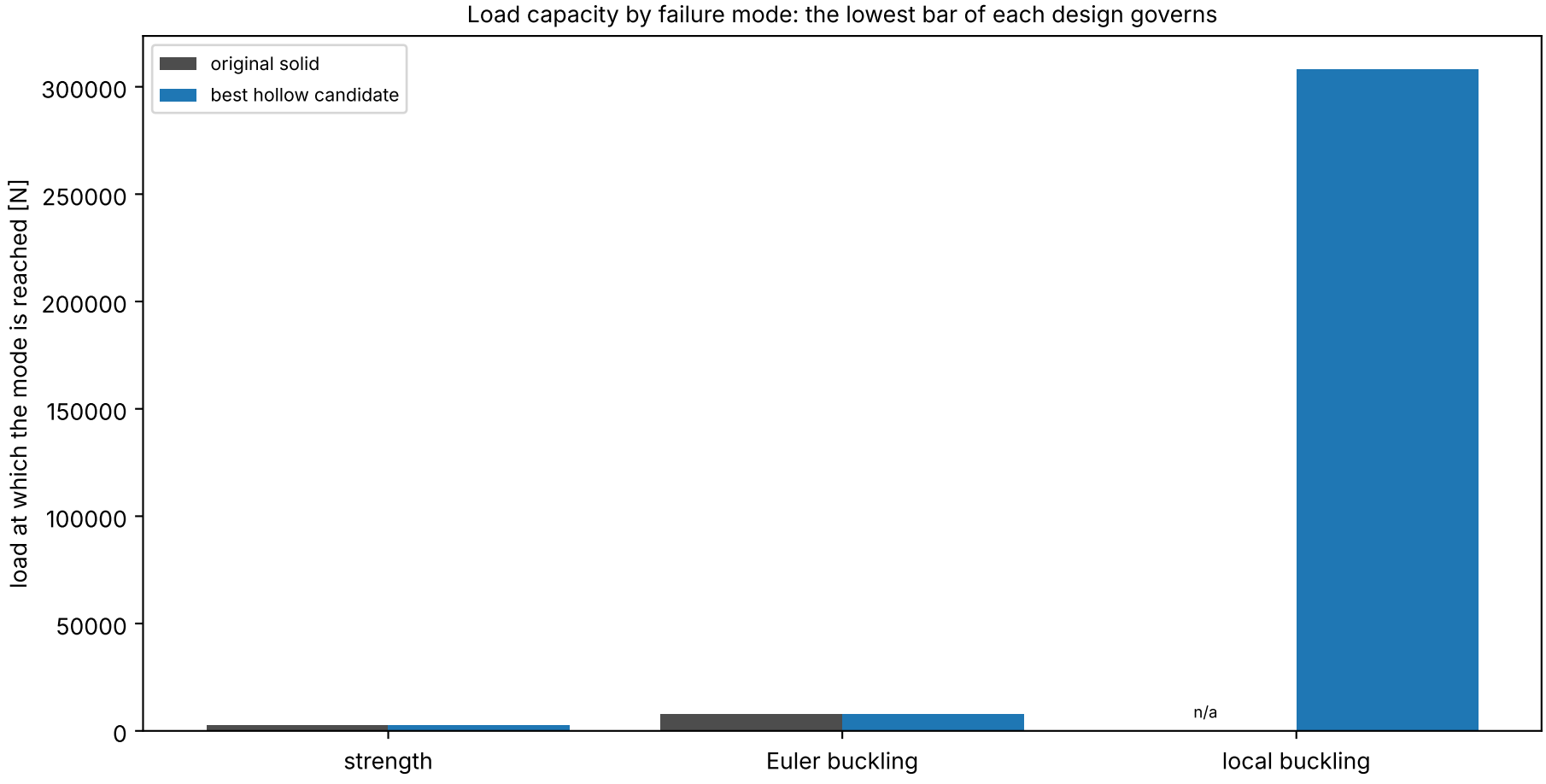

Shear-to-bending energy ratio per region (red = joint)
R = U_shear / U_bend (original solid)
share of strain energy [%] (dots)
region id, in rank order (made solid first, left)
22 11 1 21 23 2 18 12 10 9 7 6 4 3 19 20 5 8 13 27 16 25 14 28 15 17 24 26
Filled = hollow in the final design; marker area = strain-energy share
R
slenderness lambda = L / r_g (joints plotted at 2)

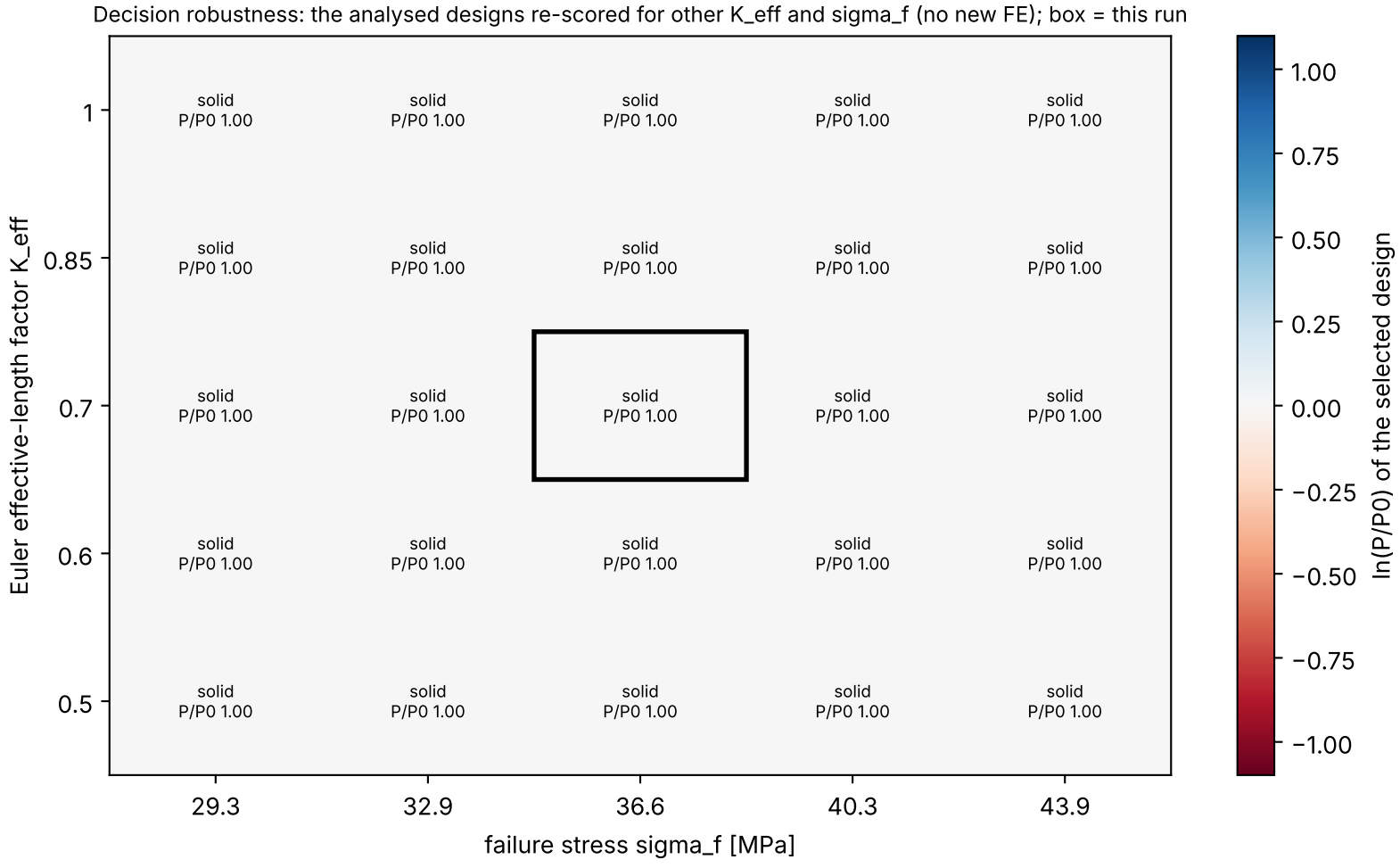

Constant-mass wall thickness for each design in the ranked family
outer offset = hollow wall s
as built (s - 0.2 mm)
t_min = 1.5 mm
analysed by FE
wall thickness [mm]
hollowed share of the part volume [%]
k = number of candidate regions kept solid (in rank order)
2.3
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
80
60
40
20
0
0
5
10
15
20
25

Regions (blue rows are hollow in the final design)

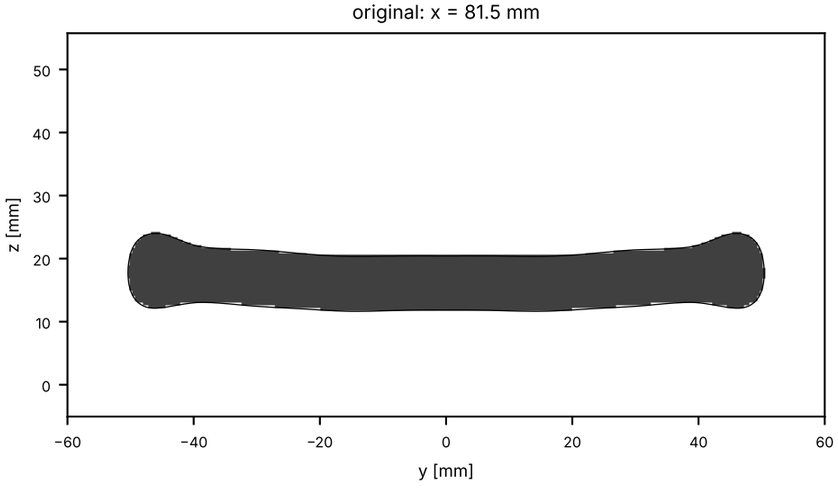
id	kind	rank	volume mm3	r mm	L mm	lambda	R	U share	final
1	joint	3	323	3	-	-	0.523	0.5%	solid
2	joint	6	298	3	-	-	0.51	0.5%	solid
3	joint	14	2.63e+03	5.1	-	-	0.147	3.0%	solid
4	joint	13	2.65e+03	5.2	-	-	0.15	3.1%	solid
5	joint	17	2.77e+03	5.39	-	-	0.126	3.0%	solid
6	joint	12	2.67e+03	5.1	-	-	0.185	2.8%	solid
7	joint	11	2.54e+03	4.9	-	-	0.199	2.7%	solid
8	joint	18	2.58e+03	4.9	-	-	0.126	2.7%	solid
9	joint	10	2.4e+03	4.47	-	-	0.218	2.7%	solid
10	joint	9	2.4e+03	4.58	-	-	0.245	2.7%	solid
11	joint	2	317	3	-	-	0.537	0.5%	solid
12	joint	8	229	3	-	-	0.377	0.3%	solid
13	strut	19	3.25e+03	5.1	44.8	17.6	0.0984	5.4%	solid
14	strut	23	3.22e+03	5	42.4	17	0.0728	5.4%	solid
15	strut	25	3.93e+03	4	72.3	36.1	0.0358	0.0%	solid
16	strut	21	3.51e+03	5.2	51	19.6	0.079	4.7%	solid
17	strut	26	3.22e+03	5	39.8	15.9	0.0334	4.4%	solid
18	strut	7	2.24e+03	3.87	45.3	23.4	0.489	0.0%	solid
19	strut	15	2.78e+03	4.12	53.4	25.9	0.144	3.4%	solid
20	strut	16	2.97e+03	4.47	55.5	24.8	0.141	3.6%	solid
21	strut	4	2.23e+03	4.12	45.9	22.3	0.523	0.0%	solid
22	strut	1	2.27e+03	4.12	47	22.8	0.567	0.0%	solid
23	strut	5	2.28e+03	4.12	46.7	22.7	0.514	0.0%	solid
24	strut	27	3.64e+03	5	44.2	17.7	0.0332	5.0%	solid
25	strut	22	3.56e+03	5.2	50.3	19.3	0.0733	4.8%	solid
26	strut	28	4.03e+03	4.12	70.8	34.4	0.0187	0.0%	solid
27	strut	20	3.16e+03	5.1	43.8	17.2	0.091	5.2%	solid
28	strut	24	3.2e+03	5	42.5	17	0.0626	5.4%	solid
29	mount	-	2.11e+03	4	-	-	0.857	5.8%	solid
30	mount	-	2.1e+03	4	-	-	0.906	5.8%	solid
31	mount	-	1.56e+03	4	-	-	0.202	2.5%	solid
32	mount	-	1.56e+03	4	-	-	0.216	2.5%	solid
33	mount	-	1.23e+03	3.46	-	-	0.849	0.0%	solid
34	mount	-	1.22e+03	3.74	-	-	0.818	0.0%	solid
35	mount	-	2.16e+03	4.12	-	-	0.824	5.8%	solid
36	mount	-	2.14e+03	4	-	-	0.718	5.7%	solid

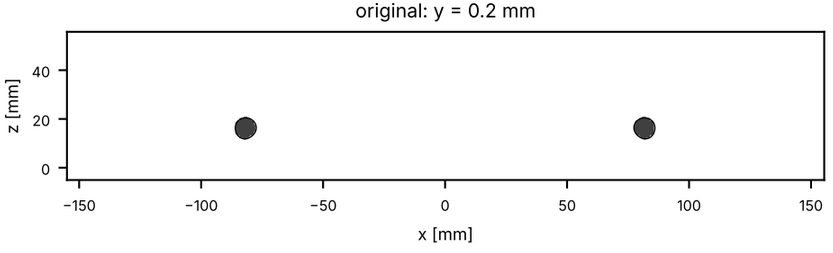

Performance of the analysed designs (governing mode: S strength, E Euler, L local)
ratio to the original solid
k (regions kept solid)
K / K0 (stiffness)
P / P0 (load capacity)
stiffness floor
capacity floor
Objective (blue = acceptable); best k = 27, rejected → solid kept
J = [w_K ln(K/K0) + w_P ln(P/P0)] / (w_K + w_P)
acceptance: ln(1 + 0.02)
k

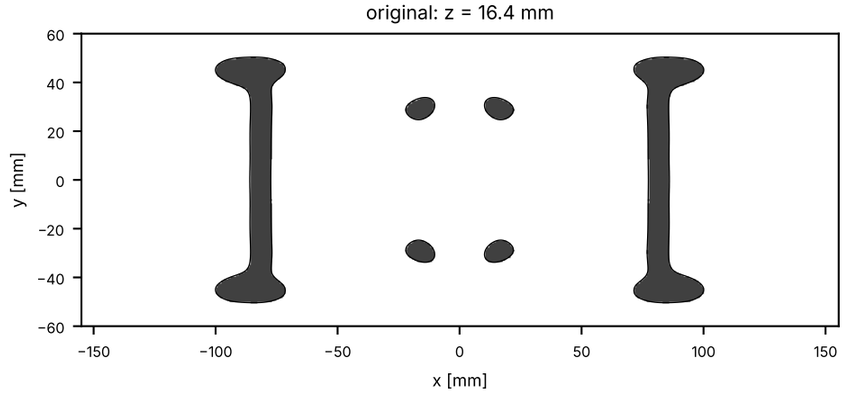

Load capacity by failure mode: the lowest bar of each design governs
original solid
best hollow candidate
load at which the mode is reached [N]
300000
250000
200000
150000
100000
50000
0
n/a
strength
Euler buckling
local buckling

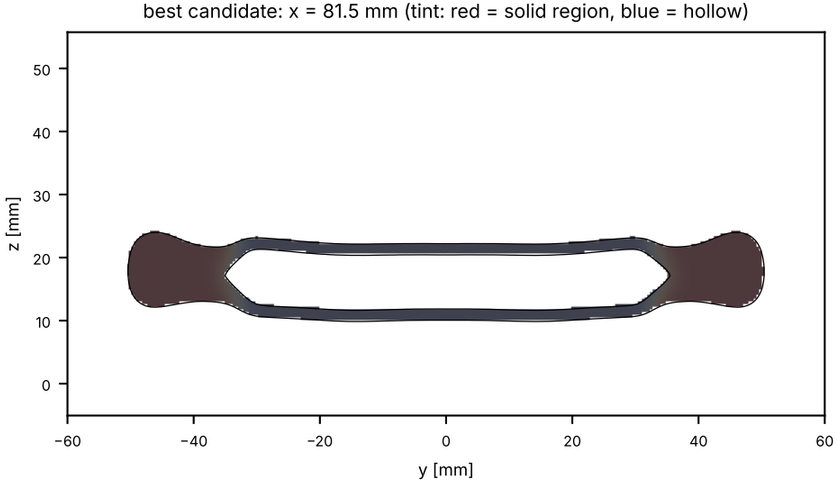

Decision robustness: the analysed designs re-scored for other K_eff and sigma_f (no new FE); box = this run
solid
P/P0 1.00
Euler effective-length factor K_eff
1
0.85
0.7
0.6
0.5
29.3
32.9
36.6
40.3
43.9
failure stress sigma_f [MPa]
ln(P/P0) of the selected design
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

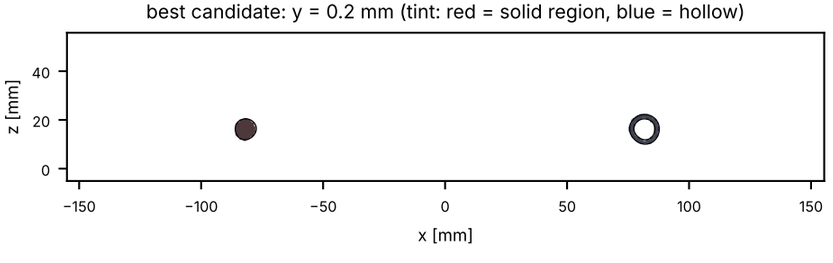
Sections through the largest hollow region of the best candidate (or the largest joint); material grey, void white

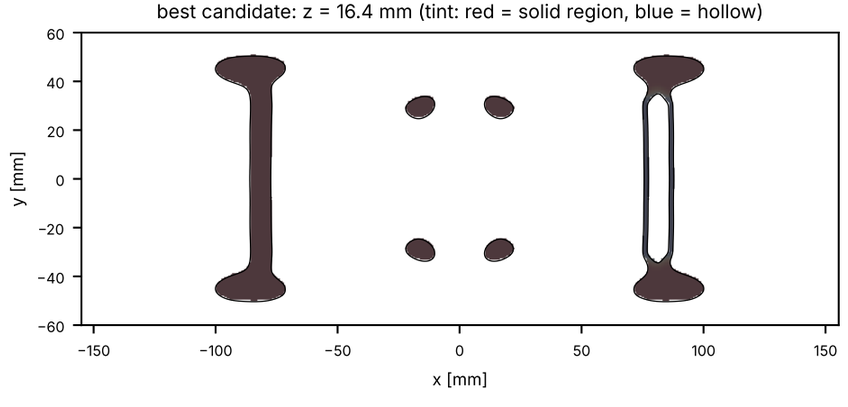




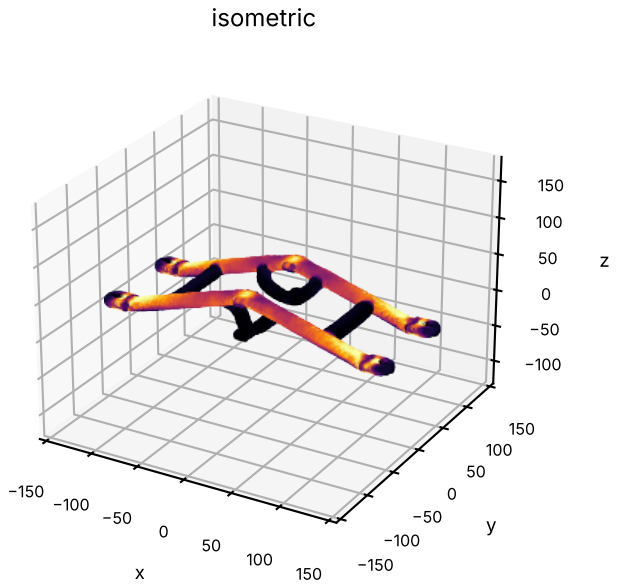


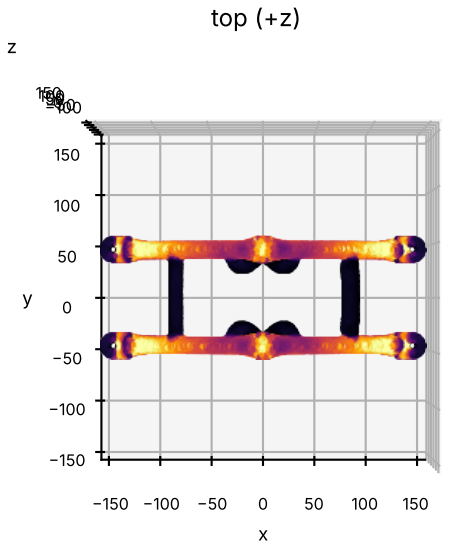


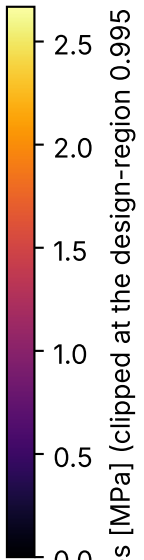


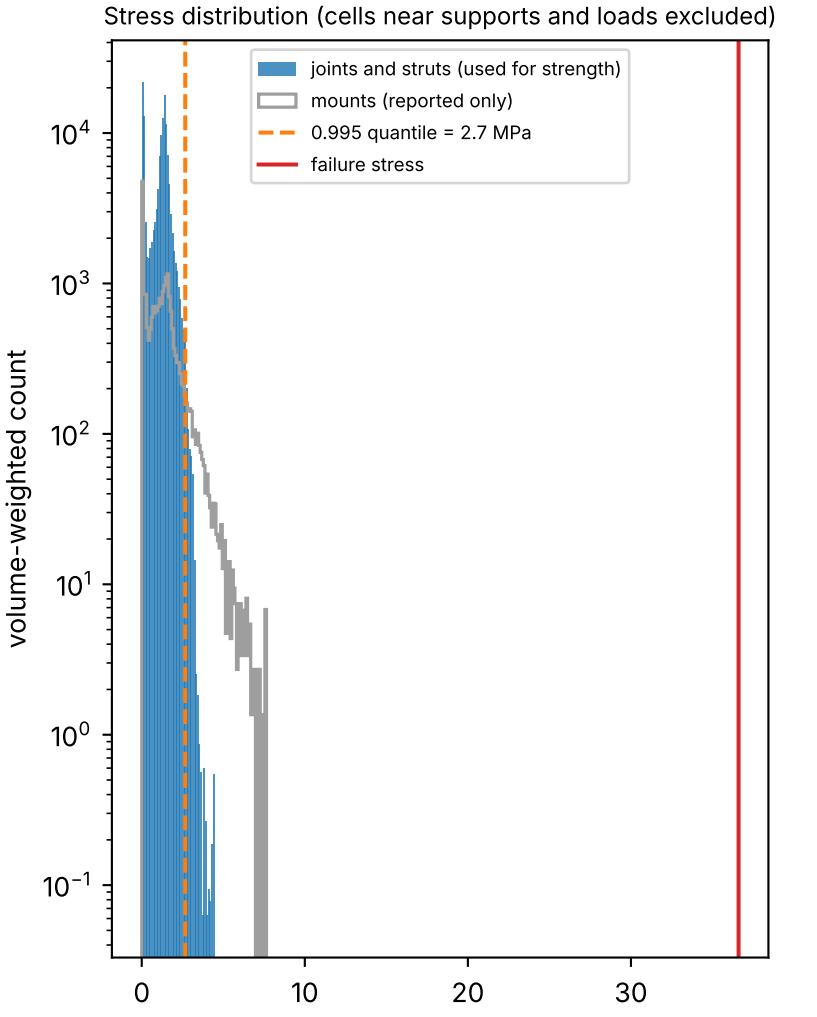


Best hollow candidate: von Mises stress at |F| = 200 N



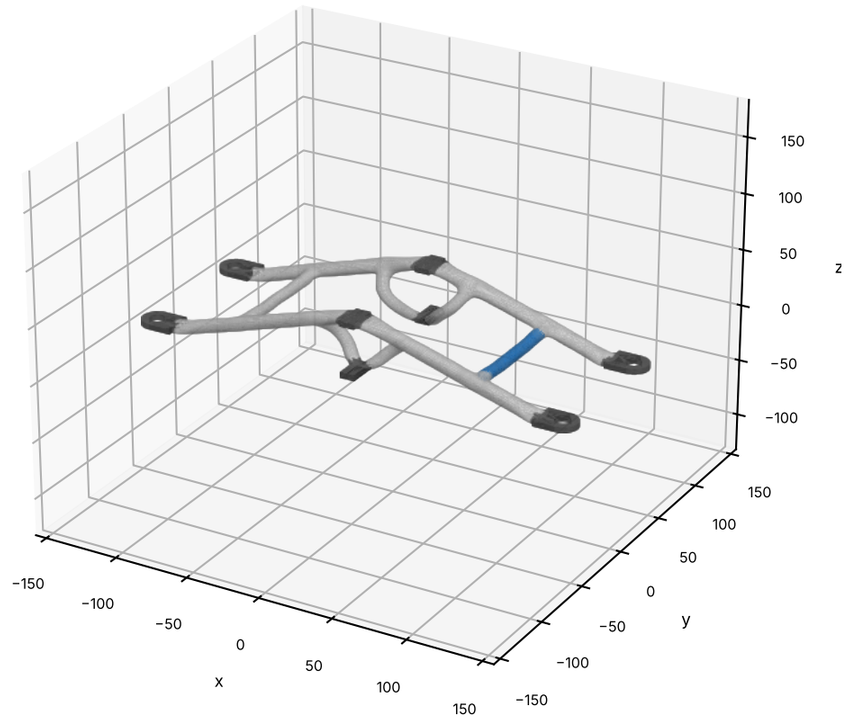




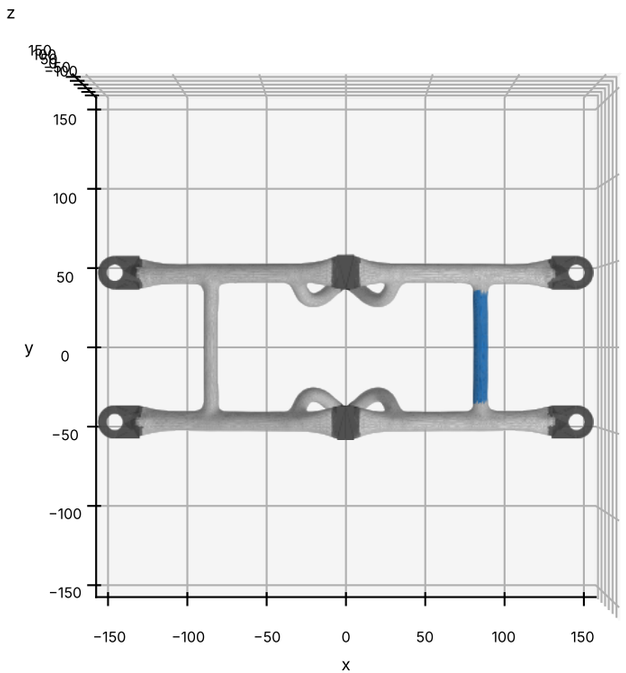


Best candidate: hollow regions blue (wall 1.77 mm), solid regions light grey, mounts dark grey - NOT adopted: final_design.stl is the original solid

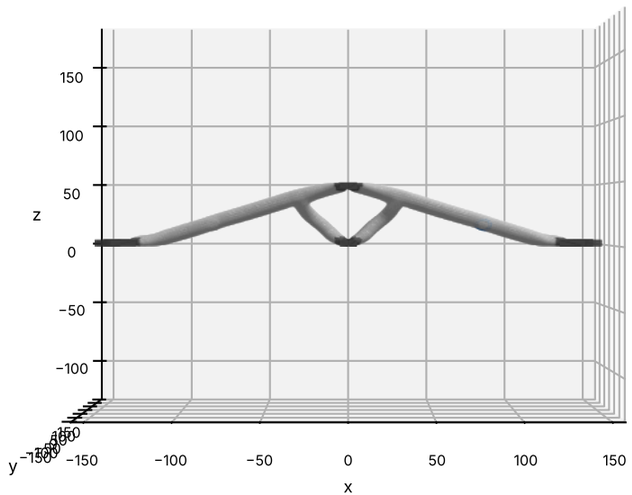
isometric



top (+z)



front (-y)

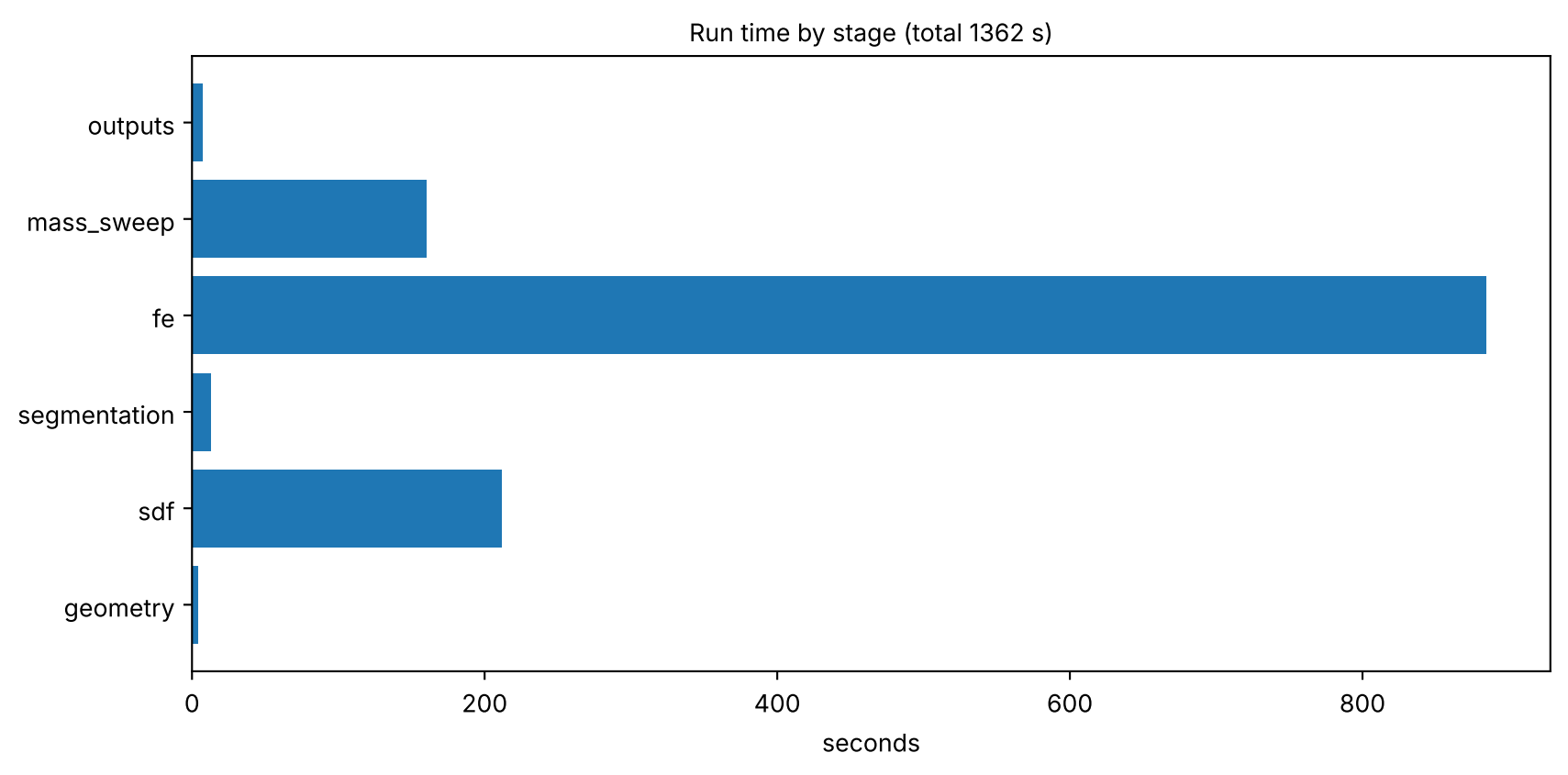


```
Enclosed voids of the best candidate: 1
  void 2: 3794 mm3

Small voids filled before analysis: 0.0 mm3

Suggested escape holes (lowest-stress wall points, outward direction n):
  void 2: (85.6, -0.2, 17.5) n = (+0.96, -0.00, +0.29), von Mises 0.0 MPa, d = 3.0 mm
  void 2: (85.2, -27.4, 18.7) n = (+0.92, +0.01, +0.40), von Mises 0.0 MPa, d = 3.0 mm

Escape holes are suggestions only: they are not cut into the mesh and their mass
(about pi d^2 t / 4 each) is not included in the constant-mass balance.
```


Run time by stage (total 1362 s)
outputs
mass_sweep
fe
segmentation
sdf
geometry
0
200
400
600
800
seconds